Google Play Games sign-in: what's left for you (owner)

The code side is finished: Play Games Services v2 SDK (`play-services-games-v2:21.0.0`) inside our CastlePlay plugin, project id `300021741882` in the manifest, automatic sign-in at start, and a Settings → Play Games → **Sign in** button. Nothing more is needed in code.

Sign-in only works when the **SHA-1 fingerprint of the key that signed the installed app** is registered as an Android credential in Play Games Services. There are three keys:

Key	SHA-1	Signs	I
App signing key (Google holds it)	copy it from Play Console (step 1)	every copy installed **from Google Play**	**al pl R**
Upload key (`~/castle_upload.keystore`)	`E8:2E:B0:4A:D6:0A:B3:C2:F8:6A:8E:68:C2:9C:97:E5:8D:CC:5A:D2`	`build/stone_storm.aab` before Google re-signs it	op (o in A yo wi bu
Debug key (`~/.android/debug.keystore`)	`55:4F:3B:3C:E4:E3:76:84:9A:35:EC:33:D6:46:D4:79:B9:67:52:A1`	`build/stone_storm_debug.apk` (friends' test-ads copy)	op (s th Al

Step 1. Copy the app signing key's SHA-1

1. Open https://play.google.com/console and pick **Stone Storm** (Castle Smashers).
2. Left menu: **Test and release** → **App integrity**.
3. Tab or section **Play app signing** → click **Settings** (or "App signing key certificate").
4. Under **App signing key certificate**, find **SHA-1 certificate fingerprint**. Click the copy icon next to it. It looks like `AB:12:...` (20 pairs). This is **not** the E8:2E... one above.

Step 2. Add it as a credential

1. Left menu: **Grow users** → **Play Games Services** → **Setup and management** → **Configuration**.
2. You should see project **300021741882** and a credential for package `com.castlesmashers.game`.
3. Click the credential (or **Add credential** → **Android**).
4. If it already lists a SHA-1, check which one. If it is the upload key (E8:2E...), add a **second** Android credential:
5. Name: `Stone Storm (Play signing)`
6. Package name: `com.castlesmashers.game`
7. **Signing certificate fingerprint (SHA-1)**: paste the value from Step 1
8. Leave "Anti-piracy" off. Click **Save changes**.
9. Optional, for the friends' APK: add one more Android credential with the debug SHA-1 `55:4F:3B:3C:E4:E3:76:84:9A:35:EC:33:D6:46:D4:79:B9:67:52:A1`. Google may warn that the package already has a credential: that's fine.

Note: if Google says the credential must be created in Google Cloud first, open the **OAuth consent / Google Cloud** link it shows, create an **Android OAuth client** with the same package and SHA-1, then come back and pick it. Use the same Google account as Play Console.

Step 3. Testers (while Play Games Services is unpublished)

1. **Play Games Services** → **Setup and management** → **Testers** → **Add testers**.
2. Add your own Gmail and anyone testing. Unpublished projects only let testers sign in.

Step 4. Test on your phone

1. Install 1.1.0 **from the Play closed testing link** (not the APK).
2. Open the game. Within a few seconds Google shows "Signed in as ..." at the top, or Settings → Play Games shows your name instead of the Sign in button.
3. If it shows **Sign in** and tapping it does nothing or flashes: the SHA-1 is wrong or the account is not a tester. Recheck Step 2 and 3.

Step 5. Publish

1. **Play Games Services** → **Publishing** (or "Review and publish").
2. It lists anything missing (for example the game's icon or description for Play Games). Fill those in: icon `assets/art/brand/play_store_icon_512.png`, description from `store_listing/listing.md`.
3. Click **Publish**. After publishing every player can sign in, not just testers.

What sign-in does today

Players get the Play Games welcome banner and their Play Games profile. **Progress is not stored in the cloud yet** (Saved Games is enabled in the console but not used by the game). Saves stay on the phone and in Android backup as before. Cloud save is not planned for now (owner's decision).